

CLASH OF T-AI-TANS

PLANT DISEASE DETECTION MODEL

```
!git clone https://github.com/pratikkayal/PlantDoc-Dataset.git
```

```
Cloning into 'PlantDoc-Dataset'...
remote: Enumerating objects: 2670, done.
remote: Counting objects: 100% (42/42), done.
remote: Compressing objects: 100% (20/20), done.
remote: Total 2670 (delta 23), reused 40 (delta 22), pack-reused 2628
Receiving objects: 100% (2670/2670), 932.92 MiB | 32.42 MiB/s, done.
Resolving deltas: 100% (24/24), done.
Updating files: 100% (2581/2581), done.
```

Thanks to Pratik Kayal for the Dataset Of Plant Leaf Images <https://github.com/pratikkayal/PlantDoc-Dataset.git>

```
import os
```

```
os.chdir('/content/PlantDoc-Dataset')
os.listdir('/content/PlantDoc-Dataset')
```

```
['logs',
 'results',
 'PlantDoc_Examples.png',
 'LICENSE.txt',
 'PlantDoc-Dataset',
 'train',
 '.git',
 'README.md',
 'test']
```

Installing Necessary Libraries For Image Classification Model

```
# Installing necessary libraries
!pip install tensorflow transformers
```

```
# Importing libraries
import tensorflow as tf
from transformers import ViTForImageClassification, ViTFeatureExtractor
from tensorflow.keras.preprocessing.image import ImageDataGenerator
import numpy as np
import matplotlib.pyplot as plt
import os
from sklearn.metrics import classification_report, confusion_matrix
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.10/dist-packages (2.17.0)
Requirement already satisfied: transformers in /usr/local/lib/python3.10/dist-packages (4.42.4)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (24.3.25)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: h5py>=3.10.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (3.11.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: ml-dtypes<0.5.0,>=0.3.1 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (0.4.0)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.10/dist-packages (from tensorflow) (24.1)
Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<5.0.0dev,>=3.20.3 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (4.21.0)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (2.32.3)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (from tensorflow) (71.0.4)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (2.4.0)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (4.12.2)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.16.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.64.1)
Requirement already satisfied: tensorboard<2.18,>=2.17 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (2.17.0)
Requirement already satisfied: keras>=3.2.0 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (3.4.1)
Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (0.37.0)
Requirement already satisfied: numpy<2.0.0,>=1.23.5 in /usr/local/lib/python3.10/dist-packages (from tensorflow) (1.26.4)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from transformers) (3.15.4)
Requirement already satisfied: huggingface-hub<1.0,>=0.23.2 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.23.5)
```

```
Requirement already satisfied: pyyaml<=5.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.10/dist-packages (from transformers) (2024.5.15)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.4.4)
Requirement already satisfied: tokenizers<0.20,>=0.19 in /usr/local/lib/python3.10/dist-packages (from transformers) (0.19.1)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.10/dist-packages (from transformers) (4.66.5)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.10/dist-packages (from astunparse>=1.6.0->tensorflow) (0.42.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.10/dist-packages (from huggingface-hub<1.0,>=0.23.2->transformers) (2024.6.1)
Requirement already satisfied: rich in /usr/local/lib/python3.10/dist-packages (from keras>=3.2.0->tensorflow) (13.7.1)
Requirement already satisfied: namex in /usr/local/lib/python3.10/dist-packages (from keras>=3.2.0->tensorflow) (0.0.8)
Requirement already satisfied: optree in /usr/local/lib/python3.10/dist-packages (from keras>=3.2.0->tensorflow) (0.12.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2.21.0->tensorflow) (2024.7.4)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.10/dist-packages (from tensorboard<2.18,>=2.17->tensorflow) (3.6.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.10/dist-packages (from tensorboard<2.18,>=2.17->tensorflow) (0.17.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from tensorboard<2.18,>=2.17->tensorflow) (3.0.6)
Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/local/lib/python3.10/dist-packages (from werkzeug>=1.0.1->tensorboard<2.18,>=2.17->tensorflow) (2.1.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.10/dist-packages (from rich->keras>=3.2.0->tensorflow) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (from rich->keras>=3.2.0->tensorflow) (2.18.0)
Requirement already satisfied: mdurl!=0.1 in /usr/local/lib/python3.10/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.2.0->tensorflow) (0.1.2)
```

✓ Image Preprocessing and Data Augmentation

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
import numpy as np
import os
import shutil

train_dir = '/content/PlantDoc-Dataset/train'
class_to_remove = 'Tomato two spotted spider mites leaf'

class_dir_to_remove = os.path.join(train_dir, class_to_remove)
if os.path.exists(class_dir_to_remove):
    shutil.rmtree(class_dir_to_remove)
    print(f'Removed class directory: {class_to_remove}')
else:
    print(f'Class directory not found: {class_to_remove}')

test_dir = '/content/PlantDoc-Dataset/test'

# Creating an ImageDataGenerator for data augmentation and normalization
train_datagen = ImageDataGenerator(rescale=1.0/255, # Normalize pixel values to [0, 1]
                                   rotation_range=20, # Randomly rotate images
                                   width_shift_range=0.2, # Randomly shift images horizontally
                                   height_shift_range=0.2, # Randomly shift images vertically
                                   shear_range=0.2, # Randomly shear images
                                   zoom_range=0.2, # Randomly zoom images
                                   horizontal_flip=True, # Randomly flip images horizontally
                                   fill_mode='nearest') # Fill missing pixels after transformations

# Loading the training data
train_generator = train_datagen.flow_from_directory(train_dir,
                                                    target_size=(224, 224),
                                                    batch_size=32,
                                                    class_mode='categorical') # Using 'categorical' for multi-class classification

# Creating a test ImageDataGenerator (only rescaling, no augmentation)
test_datagen = ImageDataGenerator(rescale=1.0/255)

test_generator = test_datagen.flow_from_directory(test_dir,
                                                  target_size=(224, 224),
                                                  batch_size=32,
                                                  class_mode='categorical',
                                                  shuffle=False) # Do not shuffle test data

📁 Class directory not found: Tomato two spotted spider mites leaf
Found 2340 images belonging to 27 classes.
Found 236 images belonging to 27 classes.
```

✓ Using a Pre Trained Model **EfficientNetB0** and fine tuning the model by adding custom layers for our problem

```

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator

base_model = tf.keras.applications.EfficientNetB0(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3)
)

x = base_model.output
x = tf.keras.layers.GlobalAveragePooling2D()(x)
x = tf.keras.layers.Dense(128, activation='relu')(x)
outputs = tf.keras.layers.Dense(train_generator.num_classes, activation='softmax')(x)

model = tf.keras.Model(inputs=base_model.input, outputs=outputs)

# Compiling the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
              loss='categorical_crossentropy',
              metrics=['accuracy'])

# Training the model
history = model.fit(train_generator,
                    validation_data=test_generator,
                    epochs=10)

```

Epoch 1/10
 /usr/local/lib/python3.10/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` c
 self._warn_if_super_not_called()
 74/74 ————— 947s 11s/step - accuracy: 0.1081 - loss: 3.2026 - val_accuracy: 0.0466 - val_loss: 3.2942
 Epoch 2/10
 74/74 ————— 878s 12s/step - accuracy: 0.3534 - loss: 2.4346 - val_accuracy: 0.0466 - val_loss: 3.2970
 Epoch 3/10
 74/74 ————— 846s 11s/step - accuracy: 0.5351 - loss: 1.6889 - val_accuracy: 0.0551 - val_loss: 3.2966
 Epoch 4/10
 74/74 ————— 890s 12s/step - accuracy: 0.6410 - loss: 1.2661 - val_accuracy: 0.0424 - val_loss: 3.2941
 Epoch 5/10
 74/74 ————— 892s 11s/step - accuracy: 0.7104 - loss: 1.0060 - val_accuracy: 0.0763 - val_loss: 3.2703
 Epoch 6/10
 74/74 ————— 861s 11s/step - accuracy: 0.7602 - loss: 0.8242 - val_accuracy: 0.2161 - val_loss: 2.9244
 Epoch 7/10
 74/74 ————— 872s 11s/step - accuracy: 0.7982 - loss: 0.6828 - val_accuracy: 0.2924 - val_loss: 2.4981
 Epoch 8/10
 74/74 ————— 841s 11s/step - accuracy: 0.8273 - loss: 0.5504 - val_accuracy: 0.3814 - val_loss: 2.0548
 Epoch 9/10
 74/74 ————— 864s 11s/step - accuracy: 0.8428 - loss: 0.5284 - val_accuracy: 0.4280 - val_loss: 1.7922
 Epoch 10/10
 74/74 ————— 860s 11s/step - accuracy: 0.8694 - loss: 0.4291 - val_accuracy: 0.5042 - val_loss: 1.5721

✓ Evaluating the model on validation data and making a confusion matrix for evaluating performance of the model!

```

val_loss, val_acc = model.evaluate(test_generator)
print(f'Validation Accuracy: {(val_acc) * 100:.2f}%')

y_true = test_generator.classes
y_pred = np.argmax(model.predict(test_generator), axis=-1)
print(classification_report(y_true, y_pred, target_names=test_generator.class_indices.keys()))

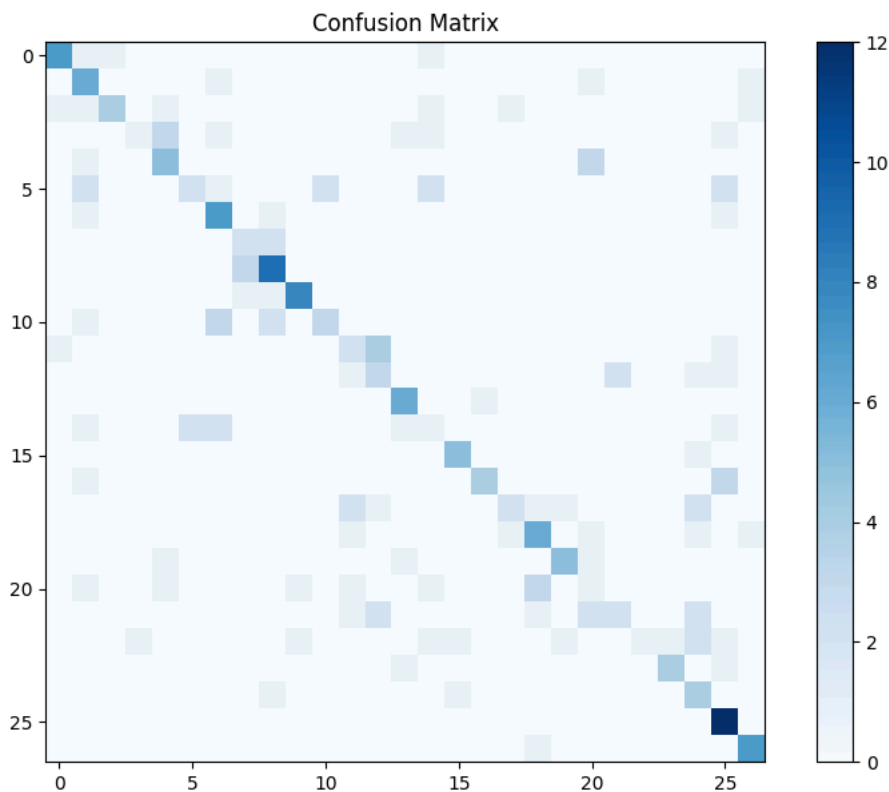
# Confusion matrix
cm = confusion_matrix(y_true, y_pred)
plt.figure(figsize=(10, 7))
plt.imshow(cm, cmap='Blues')
plt.title('Confusion Matrix')
plt.colorbar()
plt.show()

```

8/8 20s 2s/step - accuracy: 0.5012 - loss: 1.6455

Validation Accuracy: 82.42%

8/8 20s 2s/step	precision	recall	f1-score	support
Apple Scab Leaf	0.78	0.70	0.74	10
Apple leaf	0.38	0.67	0.48	9
Apple rust leaf	0.80	0.40	0.53	10
Bell_pepper leaf	0.50	0.12	0.20	8
Bell_pepper leaf spot	0.45	0.56	0.50	9
Blueberry leaf	0.50	0.18	0.27	11
Cherry leaf	0.47	0.70	0.56	10
Corn Gray leaf spot	0.33	0.50	0.40	4
Corn leaf blight	0.56	0.75	0.64	12
Corn rust leaf	0.80	0.80	0.80	10
Peach leaf	0.60	0.33	0.43	9
Potato leaf early blight	0.25	0.25	0.25	8
Potato leaf late blight	0.30	0.38	0.33	8
Raspberry leaf	0.60	0.86	0.71	7
Soyabean leaf	0.12	0.12	0.12	8
Squash Powdery mildew leaf	0.71	0.83	0.77	6
Strawberry leaf	0.80	0.50	0.62	8
Tomato Early blight leaf	0.50	0.22	0.31	9
Tomato Septoria leaf spot	0.50	0.55	0.52	11
Tomato leaf	0.71	0.62	0.67	8
Tomato leaf bacterial spot	0.11	0.11	0.11	9
Tomato leaf late blight	0.50	0.20	0.29	10
Tomato leaf mosaic virus	1.00	0.10	0.18	10
Tomato leaf yellow virus	0.80	0.67	0.73	6
Tomato mold leaf	0.31	0.67	0.42	6
grape leaf	0.50	1.00	0.67	12
grape leaf black rot	0.70	0.88	0.78	8
accuracy			0.50	236
macro avg	0.54	0.51	0.48	236
weighted avg	0.55	0.50	0.48	236



```
model_save_path = '/content/drive/My Drive/plant_disease_vit_model.h5'
model.save(model_save_path)
```

3.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my`

GRADIO INTERFACE FOR THE MODEL

```
import gradio as gr
import tensorflow as tf
import numpy as np
from tensorflow.keras.preprocessing import image
```

```
model = tf.keras.models.load_model('/content/drive/MyDrive/plant_disease_vit_model.h5')
```

```
class_names = [
    'Apple Scab Leaf', 'Apple leaf', 'Apple rust leaf',
    'Bell_pepper leaf spot', 'Bell_pepper leaf', 'Blueberry leaf',
    'Cherry leaf', 'Corn Gray leaf spot', 'Corn leaf blight',
    'Corn rust leaf', 'Peach leaf', 'Potato leaf early blight',
    'Potato leaf late blight', 'Raspberry leaf', 'Soyabean leaf',
    'Squash Powdery mildew leaf', 'Strawberry leaf', 'Tomato Early blight leaf',
    'Tomato Septoria leaf spot', 'Tomato leaf bacterial spot',
    'Tomato leaf late blight', 'Tomato leaf mosaic virus',
    'Tomato leaf yellow virus', 'Tomato leaf', 'Tomato mold leaf',
    'grape leaf black rot', 'grape leaf'
]
```

```
def predict_disease(img):
```

```
    img = img.resize((224, 224))
    img_array = image.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array = img_array / 255.0
```

```
    predictions = model.predict(img_array)
    predicted_class = np.argmax(predictions[0])
    return class_names[predicted_class]
```

```
interface = gr.Interface(
    fn=predict_disease,
    inputs=gr.Image(type="pil", label="Upload Plant Leaf Image"),
    outputs=gr.Textbox(label="Predicted Disease Class"),
    live=True
)
interface.launch()
```

 WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until , Setting queue=True in a Colab notebook requires sharing enabled. Setting `share=True` (you can turn this off by setting `share=False`

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
Running on public URL: <https://aa272b25df803af22f.gradio.live>

This share link expires in 72 hours. For free permanent hosting and GPU upgrades, run `gradio deploy` from Terminal to deploy to Sp







Clear

Predicted Disease Class

Corn Gray leaf spot

Flag

Use via API  · Built with Gradio 